

Rohit Shende

AI Engineer · ML Engineer · Python Developer

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AI Engineer and Python Developer with 3+ years of production experience building LLM-powered multi-agent systems, RAG pipelines, and REST APIs. Shipped 10+ FastAPI microservices on GCP Cloud Run. Expert in LangGraph, Gemini, CopilotKit, and n8n workflow automation. Designs end-to-end AI systems — from agent orchestration and prompt engineering to Dockerized deployment and API security hardening.

TECHNICAL SKILLS

AI / LLM: LangGraph · LangChain · Gemini · OpenAI API · LLaMA · CopilotKit · Supermemory · RAG Pipelines · Prompt Engineering

Python: FastAPI · Flask · Pydantic · Asyncio · Pandas · NumPy · Scikit-learn · PyTorch

JavaScript: Node.js · Express.js · REST APIs · Async/Await · React · npm

IoT & Embedded: MQTT · WebSockets · Modbus · RS485 · Raspberry Pi · TimescaleDB · Real-time Telemetry

Cloud & MLOps: GCP Cloud Run · Docker · Artifact Registry · Secret Manager · Cloud Armor · Firebase · App Check

Databases: MongoDB Atlas · Firestore · PostgreSQL · SQLAlchemy · Mongoose

Automation: n8n · SendGrid · WhatsApp Business API · Razorpay · Webhooks · Git

PROFESSIONAL EXPERIENCE

AI Engineer & Python Developer

Oct 2023 – Present

Grid Insight Technologies Pvt. Ltd. · Nagpur, India

- Architected a LangGraph + Gemini multi-agent system with tool-calling, conditional routing, and persistent session memory via Supermemory — deployed on GCP Cloud Run with CopilotKit UI, serving real-time energy advisory to end users.
- Engineered full prompt engineering pipeline — system prompts, few-shot examples, output schema enforcement (Pydantic), and LLM fallback chains — improving response accuracy and eliminating ~100% of unstructured outputs in production.
- Shipped 10+ FastAPI microservices containerized with Docker on GCP Cloud Run; hardened with slowapi rate limiting, Firebase App Check, and Cloud Armor; secrets managed via GCP Secret Manager — achieving zero breach incidents.
- Built RAG-style document AI pipeline for Marathi & English MSEDCL bill PDFs — structured data extraction feeding downstream LLM analytics, eliminating manual bill entry for 5,000+ monthly consumer records.
- Developed 5+ n8n automation workflows: Gmail PDF ingestion → AI extraction → MongoDB storage; WhatsApp Business API delivery; SendGrid campaigns; MongoDB → Google Sheets CRM sync — cutting manual ops time by ~80%.
- Delivered BESS & Solar financial modeling APIs (bessDetails, bessCalculatorReport, oaSolarCapacity) with LLM-generated insight commentary, enabling clients to model solar ROI and BESS savings with Excel-verified accuracy.
- Deployed IoT monitoring system across energy sites — Raspberry Pi devices ingest meter, inverter, and generation data via Modbus/RS485 and MQTT, stream to server over WebSockets, persist in TimescaleDB, and trigger real-time consumer alerts on anomalies.
- Integrated Razorpay payment gateway with first-time-user coupon engine; published AI/ML engineering content on Medium documenting production learnings.

KEY PROJECTS

AI Energy Advisor Agent | LangGraph · Gemini · CopilotKit · Supermemory · GCP Cloud Run

- Designed a stateful multi-agent system using LangGraph state machines with tool-calling and agent memory. Handles live tariff Q&A, solar sizing, and demand-charge optimization grounded on user billing data — shipped to production.

MSEDCL Bill Intelligence Pipeline | FastAPI · Python · MongoDB Atlas · n8n · WhatsApp Business API

- Engineered a fully automated AI pipeline: Gmail monitoring → PDF extraction (EN/MR) → LLM parsing → anomaly detection → WhatsApp push notification. Processes 500+ bills/month with zero manual intervention.

Volt Platform — AI-Powered Energy SaaS | FastAPI · Gemini · MongoDB · GCP · React

- Built a full-stack AI SaaS with 10+ analytics APIs; LLM commentary generation on every report output; 6-tab React dashboard with a dedicated AI Insights tab powered by Gemini — deployed on GCP Cloud Run.

Grid IoT Monitoring System | MQTT · Modbus · WebSockets · TimescaleDB · Raspberry Pi · FastAPI

- End-to-end IoT platform: Raspberry Pi devices collect meter, inverter, and solar generation data via Modbus/RS485 and MQTT, stream to FastAPI server over WebSockets, persist in TimescaleDB, and surface on a live dashboard with automated consumer anomaly alerts.

EDUCATION & CERTIFICATIONS

B.E. — Electrical & Electronics Engineering

2017 – 2021

Priyadarshini Institute of Engineering & Technology (RTMNU) · Nagpur

- ML & Data Science Bootcamp — Zero to Mastery
- PLC Programming

- PyTorch for Deep Learning — Zero to Mastery
- SCADA

Languages: English (Full Professional) · Hindi (Full Professional) · Marathi (Full Professional)